BIN DERANGEMENTS AND THE GATE WIDTH THEOREM

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Abstract. The autocorrelation $R(\ell)$ of a prime repetend vanishes at lag ℓ precisely when the corresponding group element $b^\ell \in (\mathbb{Z}/p\mathbb{Z})^*$ deranges the digit-bin partition: multiplication by b^ℓ moves every residue to a different bin. We prove that for any prime $p > b$ (no primitive-root hypothesis needed), the bin-deranging set is exactly $\{-u/(b-u) \bmod p : u = 1, \ldots, b-1\}$. In particular, there are exactly $b-1$ such elements, independent of the prime. The gate that controls which Ramanujan lags survive in the phase-filtered autocorrelation has a universal width determined solely by the base. The key observation is that the permutation $r \mapsto br \bmod p$ transforms the contiguous Beatty bins into residue classes modulo b.

For the non-deranging multipliers $g \neq 1$ (those with $C(g) > 0$), the mean collision count satisfies an exact formula:

$$\overline{R} = \frac{Q\big(b(Q-1) + 2R\big)}{b(Q-1) + R},$$

where $Q = \lfloor (p-1)/b \rfloor$ and $R = (p-1) \bmod b$. For the spectral class $R = 0$, this simplifies to $\overline{R} = Q$ exactly. For $Q = 1$ (the smallest primes $p > b+1$ in any class), the mean is always 2.

Together, these results decompose the autocorrelation into two independent controls: the base determines how many lags produce zero autocorrelation $(b-1)$, while the prime determines which lags are blocked and how strongly the surviving lags contribute.

1. Introduction

This paper characterizes the zero set of the autocorrelation.

The digit function $\delta(r) = \lfloor br/p \rfloor$, which assigns to each residue $r \bmod p$ its leading digit in base b, has been studied in connection with the distribution of digits of primes [2], exponential sums over digit constraints [4], and the statistical properties of decimal expansions [1]. The function partitions $\{1, \ldots, p-1\}$ into b bins $B_0, \ldots, B_{b-1}$, where

$$B_d = \{r \in \{1, \ldots, p-1\} : \delta(r) = d\}.$$

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These bins form a Beatty partition: the bin sizes are $\lfloor (d+1)p/b \rfloor - \lfloor dp/b \rfloor$, determined by the three-distance theorem for the sequence $\{r/p\}$ at scale $1/b$ [5].

The autocorrelation of the repetend of $1/p$ at lag ℓ counts positions j where the digit at position j equals the digit at position $j+\ell$:

$$R(\ell) = \left|\{j \in \{0, \ldots, L-1\} : \delta(b^j) = \delta(b^{j+\ell})\}\right| = \sum_{d=0}^{b-1} |B_d \cap gB_d|,$$

where $g = b^\ell \bmod p$ and the intersection is under multiplication in $(\mathbb{Z}/p\mathbb{Z})^\times$.

The question of when $R(\ell) = 0$ reduces to a combinatorial question about the Beatty partition: for which group elements g does multiplication by g move every residue to a different bin? Such elements are *derangements* of the bin partition, in the sense of the classical theory of permutation derangements [6]. We determine their exact number and give an explicit formula.

2. The Bin Partition

The bins $B_0, \ldots, B_{b-1}$ are contiguous intervals of $\{1, \ldots, p-1\}$ determined by the floor function $\delta(r) = \lfloor br/p \rfloor$. Their sizes are controlled by the division $p-1 = bQ+R$, where $Q = \lfloor (p-1)/b \rfloor$ and $R = (p-1) \bmod b$: exactly R bins have size $Q+1$ and $b-R$ bins have size Q. This distribution is a special case of the three-distance theorem [5] applied to the Farey-like partition of $[0,1)$ by the points r/p.

When $p \leq b+1$, each bin contains at most one residue, the partition is injective, and the autocorrelation is trivially zero at all non-zero lags. When $p > b+1$, multiple residues share bins, and non-trivial autocorrelation structure emerges.

3. Bin Derangements

Definition 1. An element $g \in (\mathbb{Z}/p\mathbb{Z})^*$ is a *bin-deranging multiplier* (or *bin derangement*) if

$$gB_d \cap B_d = \varnothing \quad \text{for every } d \in \{0, \ldots, b-1\}.$$

That is, multiplication by g moves every residue out of its digit bin. The terminology follows the classical notion of a derangement (a permutation with no fixed points) [6], extended to the partition setting: no bin is preserved setwise.

Proposition 2. *For a prime p where b is a primitive root, $R(\ell) = 0$ if and only if $g = b^\ell \bmod p$ is a bin-deranging multiplier.*

Proof. $R(\ell) = \sum_d |B_d \cap gB_d|$ where $g = b^\ell$. Each term is non-negative, so $R(\ell) = 0$ if and only if every term vanishes, which is the bin-derangement condition. □

The bin-derangement condition has a geometric interpretation. The bins $B_0, \ldots, B_{b-1}$ are contiguous intervals in $\mathbb{Z}/p\mathbb{Z}$ (since δ is a floor function of a linear map). Multiplication by g stretches and wraps these intervals. A bin derangement is a multiplier that stretches every interval far enough that it no longer overlaps its original position.

4. The Gate Width Theorem

Lemma 3 (Conjugation)**.** *For each $r \in \{1, \ldots, p-1\}$, let $x = [br]_p$ (the least positive residue of br modulo p). Then $br = p\,\delta(r) + x$, and consequently*

$$\delta(r) = \delta(s) \quad \Longleftrightarrow \quad [br]_p \equiv [bs]_p \pmod{b}.$$

Under the permutation $r \mapsto [br]_p$, the digit bins are exactly the residue classes modulo b.

Proof. Since $p \nmid br$, the least positive residue of br modulo p is the remainder after division: $br = p\lfloor br/p \rfloor + [br]_p = p\,\delta(r) + x$. Reducing modulo b: $0 \equiv p\,\delta(r) + x \pmod{b}$. Since $\gcd(p, b) = 1$, the residue class of x modulo b determines $\delta(r)$ uniquely. □

Lemma 4. *For every $g \in \mathbb{F}_p^\times$,*

$$C(g) = \#\{x \in \{1, \ldots, p-1\} : x \equiv [gx]_p \pmod{b}\}.$$

Proof. The map $r \mapsto x = [br]_p$ is a permutation of $\{1, \ldots, p-1\}$. Apply Lemma 3 to the pair r and gr mod p. □

Lemma 5. *Fix $g \neq 1$ and define $c \equiv b(1-g)^{-1} \pmod{p}$ with $1 \leq c \leq p-1$. If $1 \leq c \leq b-1$, then $C(g) = 0$.*

Proof. The key geometric idea is that when c is small (between 1 and $b-1$), the shifted sequence mc mod p stays too close to zero to reach the reference sequence mb, preventing any collision.

A collision gives $y = [gx]_p$ with $y = x + mb$ for some integer m, and $x \equiv -mc \pmod{p}$.

If $m \geq 0$: from $x + mb \leq p-1$, $1 \leq x + mc \leq p-1-m(b-c) < p$, but $x + mc \equiv 0 \pmod{p}$, a contradiction.

If $m < 0$: write $m = -n$ with $n \geq 1$; from $x - nb \geq 1$, $0 < x - nc = (x-nb) + n(b-c) \leq p-1$, again contradicting $x - nc \equiv 0 \pmod{p}$. □

Lemma 6. *If $c \geq b+1$, then $C(g) \geq 1$.*

Proof. Conversely, when c is large enough ($c \geq b+1$), the gap between the shifted and reference sequences is wide enough that a collision point can be constructed explicitly.

Take $x = p - c$. Then $1 \leq x \leq p-b-1$, so $y := x + b = p - c + b$ lies in $\{1, \ldots, p-1\}$ with $y \equiv x \pmod{b}$. Since $x \equiv -c \pmod{p}$ and $c(1-g) \equiv b$, we get $(1-g)x \equiv -b$, hence $gx \equiv x + b = y \pmod{p}$. So x contributes a collision. □

Theorem 7 (Gate width)**.** *Let p be prime and $2 \leq b < p$. Then*

$$\{g \in \mathbb{F}_p^\times : C(g) = 0\} = \left\{ -\frac{u}{b-u} \pmod{p} : u = 1, \ldots, b-1 \right\}.$$

In particular, there are exactly $b-1$ bin-deranging elements. No primitive-root hypothesis is needed.

Proof. By Lemmas 5 and 6, $C(g) = 0$ if and only if $c = b(1-g)^{-1} \in \{1, \ldots, b-1\}$. Solving: $g = 1 - b/c$. Setting $u = b - c$ gives $g = -u/(b-u) \pmod{p}$ for $u = 1, \ldots, b-1$. These are distinct: $-u_1/(b-u_1) \equiv -u_2/(b-u_2) \pmod{p}$ implies $b(u_1 - u_2) \equiv 0 \pmod{p}$, and since $p > b$ this gives $u_1 = u_2$. □

Corollary 8. $\left| \bigcup_{d=0}^{b-1} B_d / B_d \right| = p - b.$

Remark 9. The key step is the conjugation (Lemma 3): the permutation $r \mapsto [br]_p$ transforms the contiguous Beatty bins into residue classes modulo b. The partition geometry linearizes completely under this change of variables. The squeezing argument in Lemmas 5 and 6 then determines the zero set exactly.

Remark 10. The deranging multipliers $g = -u/(b-u) \pmod{p}$ form a rational family. For $u = b/2$ (when b is even), $g \equiv -1 \pmod{p}$, the complement element. The remaining derangers come in inverse pairs: $-u/(b-u)$ and $-(b-u)/u$ are mutual inverses modulo p.

Remark 11. The number $b-1$ depends only on the base. It does not depend on the prime, the spectral class, or the scale. The gate width is a property of the coordinate system, not of the object being observed. What varies with the prime is *which* $b-1$ lags are blocked, and *how strongly* the surviving lags contribute.

5. The Sum Rule and Mean Formula

The autocorrelation satisfies a sum rule analogous to Parseval's identity for Ramanujan sums [3], connecting the collision count to the spectral power at frequency zero.

Proposition 12 (Sum rule)**.**

$$\sum_{g\in\mathbb{F}_p^\times} C(g) = \Phi(0) = \sum_{d=0}^{b-1} n_d^2,$$

where $n_d = |B_d|$ is the size of the d-th digit bin.

Proof. $\sum_g C(g) = \sum_g \sum_r [\delta(r) = \delta(gr)] = \sum_r \sum_g [\delta(gr) = \delta(r)]$. For each r, as g ranges over $\mathbb{F}_p^\times$, the element gr ranges over all of $\mathbb{F}_p^\times$. The inner sum counts how many residues share a bin with r, which is $n_{\delta(r)}$. Summing over r: $\sum_r n_{\delta(r)} = \sum_d n_d^2$. □

The bin sizes for a prime $p > b+1$ are determined by the division $p-1 = bQ + R$, where $Q = \lfloor (p-1)/b \rfloor$ and $R = (p-1) \bmod b$. There are R bins of size $Q+1$ and $b-R$ bins of size Q. The spectral power at zero is therefore

$$\Phi(0) = R(Q+1)^2 + (b-R)Q^2 = bQ^2 + R(2Q+1).$$

Theorem 13 (Mean constructive value)**.** *The mean of $C(g)$ over the $p-b-1$ constructive multipliers (those $g \neq 1$ with $C(g) > 0$) is*

$$\overline{C} = \frac{Q\big(b(Q-1) + 2R\big)}{b(Q-1) + R}.$$

When b is a primitive root modulo p, this equals the mean of $R(\ell)$ over constructive lags.

Proof. There are $p-2$ nonzero multipliers, of which $b-1$ have $C(g) = 0$ by Theorem 7. The number of constructive multipliers is $p-2-(b-1) = p-b-1 = b(Q-1)+R$.

The sum over constructive multipliers equals $\Phi(0) - C(1) = bQ^2 + R(2Q+1) - (p-1) = bQ^2 - bQ + 2RQ = Q(b(Q-1)+2R)$.

Dividing: $\overline{C} = Q(b(Q-1)+2R)/(b(Q-1)+R)$. □

Corollary 14.
1. *For $R = 0$ (class $p \equiv 1 \pmod b$): $\overline{C} = Q$ exactly.*
2. *For $Q = 1$ (the smallest primes $p > b+1$): $\overline{C} = 2$.*
3. *As $Q \to \infty$ (large primes in any class): $\overline{C} \to Q \sim p/b$.*

Proof. (1) Set $R = 0$: the formula gives $Q \cdot b(Q-1)/b(Q-1) = Q$. (2) Set $Q = 1$: the numerator is $1 \cdot (0 + 2R) = 2R$ and the denominator is $0 + R = R$, giving 2. (3) For large Q: $Q(bQ+2R)/(bQ+R) \to Q$ since $R < b$ is bounded. □

6. Computational Verification

All results have been verified using the `nfield` software engine across multiple bases.

b	p	r	Q	R	zeros	$\overline{R}$ actual	$\overline{R}$ formula
10	17	7	1	6	9	2.00	2.00
10	61	1	6	0	9	6.00	6.00
10	97	7	9	6	9	9.63	9.63
10	131	1	13	0	9	13.00	13.00
10	179	9	17	8	9	17.81	17.81
10	193	3	19	2	9	19.21	19.21
6	11	5	1	4	5	2.00	2.00
6	41	5	6	4	5	6.24	6.24
6	89	5	14	4	5	14.11	14.11
7	17	3	2	2	6	2.50	2.50
7	41	6	5	5	6	5.42	5.42
7	97	6	13	5	6	13.11	13.11
12	17	5	1	4	11	2.00	2.00
12	67	7	5	6	11	5.17	5.17

Table 1. Gate width and mean formula verified across bases. The zero count equals $b-1$ in every case, and the mean formula is exact.

7. The Two Controls

The autocorrelation of a prime repetend is governed by two independent parameters:

(1) **The base** determines the gate width: exactly $b-1$ lags are blocked, regardless of which prime is being analyzed. This is a property of the coordinate system (the positional representation), not of the prime.

(2) **The prime** determines the gate pattern and intensity. Among the $p-b-1$ constructive multipliers, the distribution of $C(g)$ values carries the structural fingerprint of the prime. The mean intensity scales as $\overline{C} \approx Q \approx p/b$ for large primes, but the distribution around the mean encodes the bin geometry.

This decomposition separates two independent controls. The number of blocked lags is universal ($b-1$), depending only on the base. The positions of blocked lags depend on p (not just its residue class modulo b),

and the mean intensity at constructive lags has an exact closed form in (Q, R).

8. Remarks

The linearization. The proof reveals that the partition geometry, which resisted Fourier, combinatorial, and representation-theoretic approaches, linearizes completely under the change of variables $r \mapsto [br]_p$. In r-space, the bins are contiguous Beatty intervals. In $x = [br]_p$ space, the same bins are the residue classes modulo b. The collision condition $\delta(r) = \delta(gr)$ becomes the congruence $x \equiv [gx]_p \pmod{b}$, and the squeezing argument in Steps 3 and 4 completes the proof.

The explicit deranging set. The deranging multipliers $g = -u/(b-u) \pmod{p}$ for $u = 1, \ldots, b-1$ form a rational family. For $u = b/2$ (when b is even), $g \equiv -1 \pmod{p}$, the complement element. The remaining derangers come in inverse pairs: $-u/(b-u)$ and $-(b-u)/u$ are mutual inverses modulo p.

Generality. The theorem holds for all primes $p > b$, not only those where b is a primitive root. The original computational verification used the primitive-root hypothesis because the autocorrelation $R(\ell)$ is defined on the orbit of b. The algebraic proof bypasses the orbit entirely, working directly with the digit function and the collision count $C(g)$.

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